Quantum Mechanics II

LEFT – 1st Semester, 2nd Term – 2025/2026

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1 Introduction and Motivation

The description of quantum systems composed of identical particles plays a central role in quantum mechanics and underlies fundamental physical phenomena such as the Pauli exclusion principle. However, many of these effects emerge abstractly from the symmetrisation postulate, according to which the total quantum state of a system of identical particles must be symmetric under particle exchange for bosons and antisymmetric for fermions. This can make their intuitive physical interpretation challenging.

This project aims to numerically and visually explore the consequences of quantum indistinguishability in simple systems of two identical particles, either bosons or fermions, confined in external potentials. In particular, the goal is to explicitly construct symmetrised or antisymmetrised total wavefunctions.

By plotting joint probability densities and correlation functions, it becomes possible to visualise phenomena such as bosonic bunching and fermionic antibunching, as well as the influence of the spin state on the spatial structure of the system. To illustrate these phenomena, an attractive potential, the harmonic oscillator, was used.

2 Results

2.1 Distinguishable Particles

To establish a point of comparison with the classical case, we first consider two distinguishable particles, whose wavefunctions do not need to be symmetrised. As expected, the resulting joint probability density is shown below.

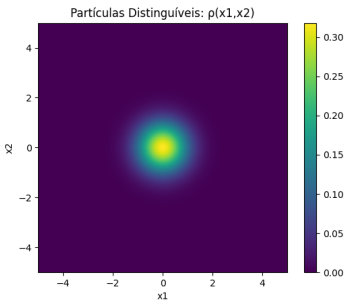


Figure 1: Joint Probability Density for Distinguishable Particles

As expected, the particle density is highest near the centre and decreases radially. We also calculated the second-order correlation function to examine the relationship between the positions of the two particles.

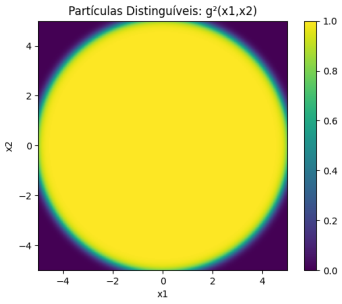


Figure 2: Correlation Function for Distinguishable Particles

The plot above shows a uniform distribution with $g^{(2)} = 1$, indicating that the two particles are uncorrelated, as expected for independent particles in the classical case.

2.2 Bosons

Similarly, we considered two spin-0 bosons in a harmonic oscillator and performed the same analysis as in the previous case. As mentioned above, these particles obey Bose–Einstein statistics, so their total wavefunction must be symmetric under particle exchange. The resulting joint probability density is shown below.

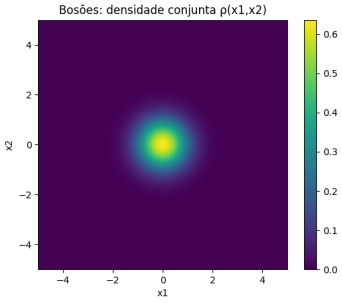


Figure 3: Ground-State Probability Density for Bosons

As can be seen, this result coincides with the classical case because both particles occupy the same spatial state. In this case, statistical correlation effects are not visible, with the only difference being the normalisation.

2.3 Fermions (Triplet State)

We also considered two indistinguishable spin-1/2 fermions. These particles obey Fermi–Dirac statistics, meaning that their total wavefunction must be antisymmetric.

It is important to distinguish between the two possible total spin values, $S = 0$ and $S = 1$, since they have different symmetry properties.

For the triplet state ($S = 1$), the spin state is symmetric, so the spatial wavefunction must be antisymmetric.

Figure 4 shows the joint probability density for the ground state ($n_1 = 0$, $n_2 = 1$).

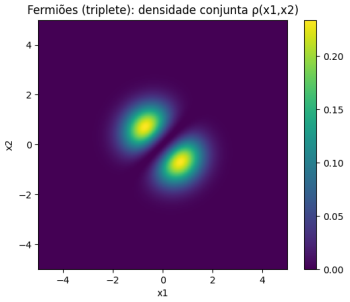


Figure 4: Ground-State Probability Density for Fermions (Triplet State)

An important feature is that the probability density vanishes along the diagonal $x_1 = x_2$. This behaviour follows from the antisymmetry of the spatial wavefunction: the two fermions cannot occupy the same single-particle quantum state, in accordance with the Pauli exclusion principle. This phenomenon is known as fermionic antibunching.

The same effect is evident in the correlation function, which also exhibits a minimum at $x_1 = x_2$, as shown in Figure 5.

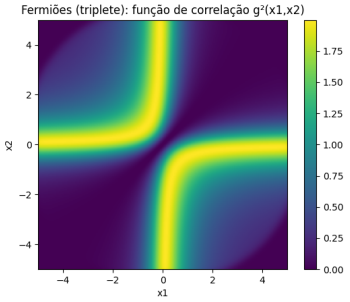


Figure 5: Ground-State Correlation Function for Fermions (Triplet State)

2.4 Fermions (Singlet State)

For the other possible total spin state of two spin-1/2 fermions, $S = 0$, the observed behaviour is similar to that of bosons. This occurs because the spin state is antisymmetric. To satisfy the spin-statistics connection, the spatial wavefunction must therefore be symmetric, as in the spin-0 bosonic case discussed above.

As an example, Figure 6 shows the probability density for $n_1 = 1$ and $n_2 = 3$.

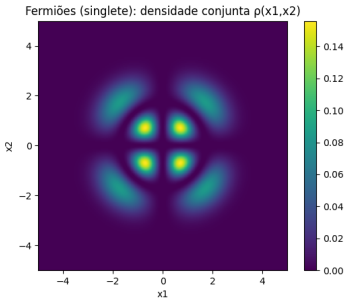


Figure 6: Probability Density for $n_1 = 1$, $n_2 = 3$ (Fermions, Singlet State)

2.5 Expectation Values

To make these effects even clearer, we calculated the expectation values of several operators. The results are presented in Table 1.

State	$\langle x_1^2 \rangle$	$\langle x_1 x_2 \rangle$	$\langle (x_1 - x_2)^2 \rangle$
Bosons	1	0	2
Fermions (Triplet)	1	-0.5	3

Table 1: Expectation values. All values are dimensionless, with $\hbar = m = \omega = 1$.

These values clearly illustrate fermionic antibunching and bosonic bunching. For example, the expectation value $\langle (x_1 - x_2)^2 \rangle$ is larger for fermions in the triplet state than for bosons, indicating a greater average spatial separation between the particles.

2.6 Infinite Potential Well

As a further investigation, we replaced the harmonic oscillator potential with an infinite potential well. Figure 7 shows the ground-state probability density for two indistinguishable fermions in the triplet spin state.

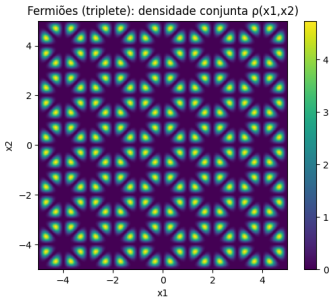


Figure 7: Ground-State Probability Density for Fermions in an Infinite Potential Well

3 Conclusion

This project presents a computational study of quantum systems composed of two identical particles, with the aim of graphically analysing the effects of quantum-state symmetrisation. The explicit construction of total wavefunctions demonstrates how quantum statistics alone can generate nontrivial spatial correlations, even in the absence of interactions.

The results clearly illustrate the differences between bosonic and fermionic behaviour: bosons can exhibit an enhanced probability of joint detection (bunching), whereas fermions can exhibit antibunching, reflecting the Pauli exclusion principle. Furthermore, the study highlights the crucial role of spin in determining the symmetry of the spatial component of the wavefunction for fermions.

This work reinforces the importance of computational tools and graphical visualisation in understanding fundamental concepts in quantum mechanics.